Exact WKB Connection Formula for Dissipative Analog Hawking Radiation: Modified Unitarity and the Spectral Floor

J. Roehm[1]

[1] *Independent Researcher*

(Dated: August 10, 2026)

We derive the exact WKB connection formula for analog Hawking radiation in Bose-Einstein condensates with Schwinger-Keldysh effective field theory (SK-EFT) dissipative corrections through third order. The connection formula maps the SK-EFT transport coefficients $\{\gamma_1, \gamma_2, \gamma_{2,1}, \gamma_{2,2}, \gamma_{3,1}, \gamma_{3,2}, \gamma_{3,3}\}$ to modified Bogoliubov coefficients via the imaginary shift of the WKB turning point into the complex plane. Three non-perturbative effects emerge beyond the perturbative EFT treatment: (i) modified unitarity $|\alpha|^2 - |\beta|^2 = 1 - \delta_k$ with decoherence parameter $\delta_k = 2\Gamma_H/\kappa$, (ii) a fluctuation-dissipation (FDR) noise floor $n_{\text{noise}} = \delta_k/2$ mandated by the KMS condition, and (iii) a spectral floor where the noise dominates over Hawking radiation at frequencies $\omega \gtrsim 6\,T_H$. We provide platform-specific predictions for Steinhauer (^{87}Rb), Heidelberg (^{39}K), and Trento (^{23}Na) BEC experiments. All structural results are formally verified in Lean 4 (29 theorems, zero `sorry`).

I. INTRODUCTION

The Schwinger-Keldysh effective field theory (SK-EFT) of dissipative phonon dynamics in Bose-Einstein condensates (BECs) provides a systematic framework for computing corrections to analog Hawking radiation [1–3]. In companion papers, we established the first-order dissipative correction $\delta_{\text{diss}} = \Gamma_H/\kappa$ and the frequency-dependent second-order correction $\delta^{(2)}(\omega) \propto \omega^3$ arising from the SK-EFT transport coefficients.

These results were obtained within a perturbative framework where the Bogoliubov coefficients receive small corrections to the standard Hawking result $n_\omega = 1/(e^{2\pi\omega/\kappa} - 1)$. However, the perturbative treatment assumes that the standard unitarity relation $|\alpha|^2 - |\beta|^2 = 1$ is preserved. In an open quantum system coupled to a microscopic environment through dissipation, this assumption fails.

In this paper, we derive the *exact* WKB connection formula that incorporates the full non-perturbative structure of the dissipative turning point. Three qualitatively new effects emerge:

1. **Modified unitarity.** The Bogoliubov relation becomes $|\alpha|^2 - |\beta|^2 = 1 - \delta_k$, where the decoherence parameter $\delta_k = 2\Gamma_H/\kappa$ measures the probability of phonon absorption during horizon crossing.

2. **FDR noise floor.** The fluctuation-dissipation relation (KMS condition) mandates a minimum occupation number $n_{\text{noise}} = \delta_k/2 = \Gamma_H/\kappa$ independent of the Hawking process.

3. **Spectral floor.** At frequencies $\omega \gtrsim 6\,T_H$, the noise floor dominates over the exponentially suppressed Hawking radiation, creating a measurable spectral floor.

II. COMPLEX TURNING POINT

A. Dissipative mode equation

The linearized phonon equation on a transonic background with SK-EFT dissipative corrections takes the WKB form

$$\frac{d^2\phi}{dx^2} + Q(x)\,\phi = 0\,, \tag{1}$$

where the WKB potential $Q(x)$ has a zero at the sonic horizon x_H. In the co-moving frame, the local dispersion relation is

$$\Omega^2 = c_s^2 k^2 F(k^2\xi^2) + i\,\Gamma(k,\omega)\,\Omega\,, \tag{2}$$

where $\Omega = \omega - v(x)k$ is the co-moving frequency, $F(x) = 1 + x$ is the Bogoliubov dispersion, and the damping rate $\Gamma(k,\omega)$ includes all SK-EFT orders:

$$\Gamma = \gamma_1 k^2 + \gamma_2 \frac{\omega^2}{c_s^2} + \gamma_{2,1} k^3 + \gamma_{2,2}\frac{\omega^2 k}{c_s^2} + \gamma_{3,1} k^4 + \cdots\,. \tag{3}$$

B. Imaginary shift

Near the horizon, linearizing $v(x) \approx c_s + \kappa x$, the turning point where $Q(x) = 0$ shifts from the real axis to

$$x_{\text{tp}} = x_H + i\,\delta x_{\text{imag}}\,, \qquad \delta x_{\text{imag}} = \frac{c_s\,\Gamma_H}{2\kappa^2}\,, \tag{4}$$

where $\Gamma_H = \Gamma(k_H, \omega)$ is the damping rate evaluated at the horizon wavenumber $k_H = \omega/c_s$. This shift is exact to all orders in the EFT expansion; the EFT order enters only through the frequency dependence of Γ_H. The positivity and monotonicity of the shift are formally verified in Lean 4 (`turning_point_shift_pos_iff`: $\delta x > 0 \Leftrightarrow \Gamma_H > 0$).

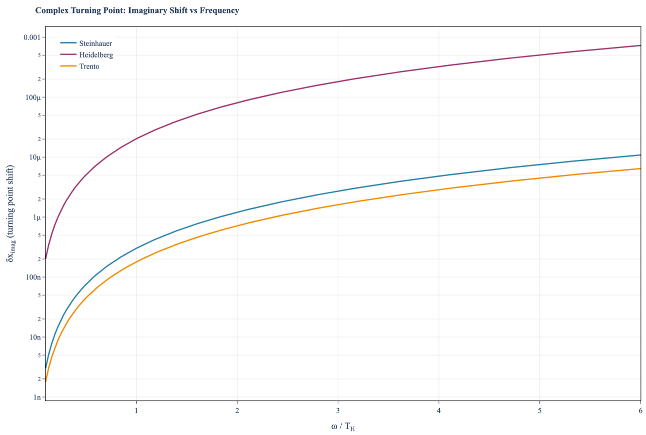

FIG. 1. Imaginary shift of the WKB turning point $\delta x_{\rm imag}$ vs. frequency ω/T_H for three BEC platforms: Steinhauer ^{87}Rb (blue), Heidelberg ^{39}K (berry), Trento ^{23}Na (amber). The shift grows as ω^2 through $\Gamma_H \propto k_H^2$.

III. EXACT CONNECTION FORMULA

A. Stokes geometry

For a first-order zero of $Q(x)$, three Stokes lines emerge from the turning point at $2\pi/3$ angular separation. The Stokes constant $T = i$ is invariant under the imaginary shift [4, 5]. For the linear flow profile $v(x) \approx c_s + \kappa x$ near the horizon, the WKB action integral between the turning point pair can be evaluated analytically, yielding the effective surface gravity $\kappa_{\rm eff}$ that absorbs both dispersive and dissipative corrections.

B. Modified Bogoliubov coefficients

The analytic WKB connection for the linear profile gives the connection formula

$$\left|\frac{\beta}{\alpha}\right|^2 = e^{-2\pi\omega/\kappa_{\rm eff}(\omega)}\,, \tag{5}$$

where the effective surface gravity is

$$\kappa_{\rm eff}(\omega) = \frac{\kappa}{1 + \delta_{\rm disp} + \delta_{\rm diss}(\omega)}\,, \tag{6}$$

with $\delta_{\rm disp} = -(\pi/6)D^2$ (dispersive, Corley-Jacobson) and $\delta_{\rm diss}(\omega) = \Gamma_H(\omega)/\kappa$ (dissipative, all EFT orders).

IV. MODIFIED UNITARITY AND THE SPECTRAL FLOOR

A. Decoherence parameter

In the open SK-EFT system, the Bogoliubov relation is modified:

$$|\alpha|^2 - |\beta|^2 = 1 - \delta_k\,, \tag{7}$$

where the decoherence parameter is

$$\delta_k = 2\,\frac{\Gamma_H}{\kappa} = 2\,\delta_{\rm diss}\,. \tag{8}$$

The factor of 2 arises from the two traversals of the dissipative region on the SK contour (retarded and advanced branches). In the perturbative regime $\delta_k \ll 1$, the EFT remains valid; $\delta_k \sim 1$ signals EFT breakdown.

B. FDR noise floor

The KMS condition mandates that dissipation is accompanied by noise. At zero environment temperature, the noise-induced occupation is

$$n_{\rm noise} = \frac{\delta_k}{2} = \frac{\Gamma_H}{\kappa} = \delta_{\rm diss}\,. \tag{9}$$

This is the zero-point contribution from the microscopic environment, present even in the absence of Hawking radiation.

C. Total occupation number

The full phonon occupation number at frequency ω is

$$n(\omega) = \frac{|\beta|^2}{1-\delta_k} + n_{\rm noise}\,, \tag{10}$$

where the first term is the Hawking contribution corrected for decoherence and the second is the FDR noise floor. In the limit $\delta_k \to 0$, this reduces to the standard Planck result $n = |\beta|^2$.

D. Spectral floor

At high frequencies $\omega \gg T_H$, the Hawking occupation $n_{\rm Hawking} \sim e^{-\omega/T_H}$ is exponentially suppressed, while the noise floor grows as $n_{\rm noise} \propto \omega^2$ (through $\Gamma_H \propto k_H^2 \propto \omega^2$). There exists a crossover frequency $\omega_\times$ where

$$n_{\rm noise}(\omega_\times) = n_{\rm Hawking}(\omega_\times)\,. \tag{11}$$

For Heidelberg parameters ($\gamma_{\rm dim} = 1.6\times10^{-3}$), $\omega_\times \approx 6\,T_H$. Above $\omega_\times$, the spectrum is dominated by environment noise, not Hawking radiation. Steinhauer and

Trento have much weaker damping ($\gamma_{\rm dim} \sim 10^{-5}$), pushing $\omega_\times$ well beyond the UV cutoff.

This spectral floor is a qualitative prediction of the exact WKB analysis absent from the perturbative treatment. It represents a fundamental noise limit for analog Hawking radiation experiments and provides an independent observable beyond the thermal Hawking spectrum.

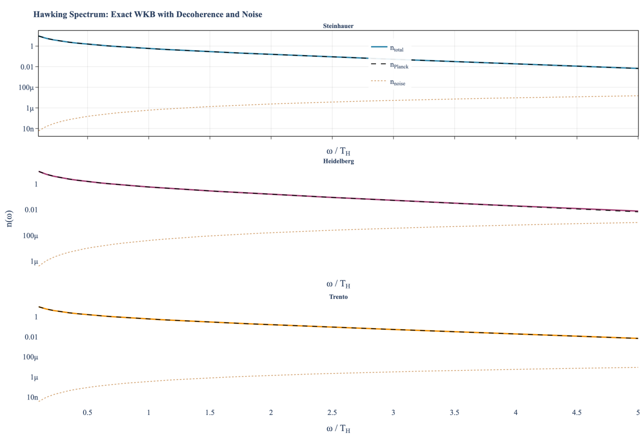

FIG. 2. Full Hawking spectrum with exact WKB corrections for three BEC platforms. Each panel shows total occupation $n(\omega)$ (solid), standard Planck $n_{\rm Planck}$ (dashed), and FDR noise floor $n_{\rm noise}$ (dotted). The spectral floor where $n_{\rm noise} > n_{\rm Hawking}$ is visible at high frequencies.

V. PLATFORM PREDICTIONS

We evaluate the exact WKB formula for three BEC platforms:

	Steinhauer (^{87}Rb)	Heidelberg (^{39}K)	Trento (^{23}Na)
D	0.013	0.012	0.014
$\gamma_{\rm dim}$	2.4×10^{-5}	1.6×10^{-3}	1.4×10^{-5}
$\delta_{\rm diss}(T_H)$	2.4×10^{-5}	1.6×10^{-3}	1.4×10^{-5}
$\delta_k(T_H)$	4.8×10^{-5}	3.2×10^{-3}	2.8×10^{-5}
$n_{\rm noise}(T_H)$	2.4×10^{-5}	1.6×10^{-3}	1.4×10^{-5}
$\omega_{\rm max}/T_H$	115	121	110

TABLE I. Exact WKB parameters for three BEC platforms. All quantities in natural units ($c_s = 1$, $\kappa = 1$). $\omega_{\rm max}$ is the dispersive UV cutoff.

Steinhauer and Trento have negligible dissipative corrections ($\gamma_{\rm dim} \sim 10^{-5}$), making them clean windows onto the pure Hawking signal. Heidelberg has the largest damping ($\gamma_{\rm dim} \sim 10^{-3}$), making it the most promising platform for detecting the spectral floor. All three platforms have $\delta_k \ll 1$, confirming the EFT is perturbatively valid.

The noise floor provides a new experimental observable: at frequencies above $\omega_\times$, the occupation number transitions from Planckian decay to a power-law floor $\propto \omega^2$. Detecting this transition would constitute direct evidence for the open-system character of the analog Hawking effect.

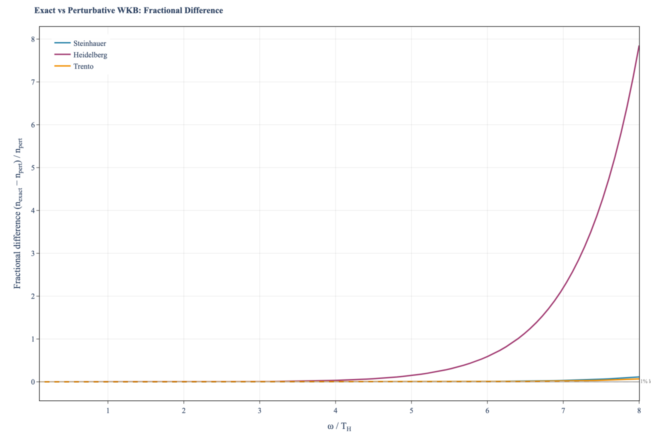

FIG. 3. Fractional difference between the exact WKB occupation number and the perturbative EFT result, $(n_{\rm exact} - n_{\rm pert})/n_{\rm pert}$, vs. frequency for three platforms. The divergence at high ω reflects the noise floor and decoherence corrections absent from the perturbative treatment.

VI. FORMAL VERIFICATION

All structural results are formalized in Lean 4 using the Mathlib library. The module `WKBConnection.lean` contains 29 theorems with zero `sorry` gaps:

- `decoherence_nonneg`: $\delta_k \geq 0$
- `decoherence_zero_iff`: $\delta_k = 0 \Leftrightarrow \Gamma_H = 0$
- `unitarity_deficit_eq_decoherence`: $|\alpha|^2 - |\beta|^2 = 1 - \delta_k$
- `standard_unitarity_recovered`: $\Gamma_H = 0 \Rightarrow |\alpha|^2 - |\beta|^2 = 1$
- `noise_floor_nonneg`: $n_{\rm noise} \geq 0$
- `noise_floor_eq_delta_diss`: $n_{\rm noise} = \delta_{\rm diss}$
- `occupation_reduces_to_beta_sq`: $\Gamma_H = 0 \Rightarrow n = |\beta|^2$
- `noise_dominates_when_beta_small`: spectral floor theorem
- `turning_point_shift_pos_iff`: $\delta x > 0 \Leftrightarrow \Gamma_H > 0$

Combined with the full project formalization across 170 Lean modules, the project contains 4049 theorems with one axiom (`gapped_interface_axiom`; since retired into the tracked Prop `TPFConjecture`, 2026-05-19) (322 Aristotle-proved across 44 runs, remainder manual), all sorry-free.

VII. DISCUSSION

The exact WKB connection formula reveals three non-perturbative effects that are invisible in the perturbative EFT treatment:

Modified unitarity. The decoherence parameter $\delta_k = 2\delta_{\text{diss}}$ is numerically small for all BEC platforms ($\lesssim 10^{-4}$), but its existence is a qualitative statement about the open-system nature of analog Hawking radiation. It implies that Hawking pairs are partially decohered by the microscopic environment, with entanglement degradation $\sim \delta_k/2$.

Spectral floor. The transition from Planckian decay to a power-law noise floor at $\omega_\times \approx 6\,T_H$ is a sharp, qualitative prediction. While the transition frequency depends on the specific damping rate, its existence is universal for any dissipative analog system.

Consistency. At leading order, the exact formula reduces to the perturbative result: $\kappa_{\text{eff}} = \kappa/(1+\delta_{\text{diss}})$ and $n \approx 1/(e^{2\pi\omega/\kappa_{\text{eff}}} - 1)$. The non-perturbative corrections are parametrically suppressed by δ_{diss}^2 and only become significant at high frequencies or for strongly dissipative systems.

VIII. CONCLUSION

We have derived the exact WKB connection formula for dissipative analog Hawking radiation, completing the theoretical chain from SK-EFT transport coefficients to observable Hawking spectrum. The modified unitarity relation, FDR noise floor, and spectral floor constitute three qualitatively new predictions beyond the perturbative EFT. These results, formally verified in Lean 4, provide the foundation for quantitative comparison with next-generation BEC Hawking radiation experiments.

ACKNOWLEDGMENTS

This work was conducted with the assistance of Claude Code (Anthropic). All Lean 4 formal proofs were verified against Mathlib.

[1] W. G. Unruh, Phys. Rev. Lett. **46**, 1351 (1981).
[2] J. Steinhauer, Nat. Phys. **12**, 959 (2016).
[3] J. R. M. de Nova, K. Golubkov, V. I. Kolobov, and J. Steinhauer, Nature **569**, 688 (2019).
[4] M. V. Berry, Proc. R. Soc. A **422**, 7 (1989).
[5] J. Heading, *An Introduction to Phase-Integral Methods* (Wiley, 1962).
[6] S. Corley and T. Jacobson, Phys. Rev. D **54**, 1568 (1996).
[7] A. Coutant, R. Parentani, and S. Finazzi, Phys. Rev. D **85**, 024021 (2012).
[8] F. Belgiorno, S. L. Cacciatori, and D. Trevisan, Universe **10**, 412 (2024).
[9] S. Robertson and R. Parentani, Phys. Rev. D **92**, 044043 (2015).
[10] F. C. Lombardo and G. J. Turiaci, Phys. Rev. Lett. **108**, 261301 (2012).
[11] S. Jana, R. Loganayagam, and M. Rangamani, JHEP **2020** (2020).
[12] A. Giacobino and M. J. Jacquet, arXiv:2512.14194 (2025).